

Logistics Regression

Sunday, April 12, 2026 4:34 PM

$y=mx+b$

- **m** → slope (how much y changes when x increases)
- **b** → intercept

Formula to Calculate m

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$
 X AND Y HEAD IS MEAN

Step-by-Step Example (Manual Calculation)

Let's take a small dataset:

X = [1, 2, 3, 4]
Y = [2, 4, 5, 8]

$$= \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$
 MEAN OF X AND Y IS 2.5 AND 4.75

Step 2: Create Table					
x	y	x - $\bar{x}$	y - $\bar{y}$	(x - $\bar{x}$)(y - $\bar{y}$)	(x - $\bar{x}$) ²
1	2	-1.5	-2.75	4.125	2.25
2	4	-0.5	-0.75	0.375	0.25
3	5	0.5	0.25	0.125	0.25
4	8	1.5	3.25	4.875	2.25

Step 1: Find Mean

$$\bar{x} = \frac{1 + 2 + 3 + 4}{4} = 2.5$$
$$\bar{y} = \frac{2 + 4 + 5 + 8}{4} = 4.75$$

Step 3: Sum Values

$$\sum (x - \bar{x})(y - \bar{y}) = 4.125 + 0.375 + 0.125 + 4.875 = 9.5$$
$$\sum (x - \bar{x})^2 = 2.25 + 0.25 + 0.25 + 2.25 = 5$$

Calculate m

$$m = \frac{9.5}{5} = 1.9$$

Final Answer

🔗 Slope (m) = 1.9

Calculate b (Intercept)

Formula:

$$b = \bar{y} - m\bar{x}$$

$$b = 4.75 - (1.9 \times 2.5) = 4.75 - 4.75 = 0$$

Final Equation

$y = 1.9x + 0$ (We find the value of m and b so we can predict the values based on X values)

Now Compare Y and Y_Predict			
x	y	Y_predict	Error in Prediction
1	2	1.9	5%
2	4	3.8	5%
3	5	5.7	-14%

Y_Predict= mx+b

4	8	7.6	5%
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Calculate Overall Error (MSE)

To evaluate the model, we use:

$$MSE = \frac{1}{n} \sum (y - \hat{y})^2$$

x	y	y_Predict	Error in Prediction	Error (Y-Y_Predict)	Error Square	Mean MSE
1	2	1.9	5%	0.1	0.01	0.175
2	4	3.8	5%	0.2	0.04	
3	5	5.7	-14%	-0.7	0.49	
4	8	7.6	5%	0.4	0.16	

Goal of Gradient Descent

We want to find **m** and **b** such that error is minimum.

🔗 Error function = MSE:

$$MSE = \frac{1}{n} \sum (y - (mx + b))^2$$

Update Rules (Core Formulas)

We update **m** and **b** repeatedly:

◆ **Slope update:**

$$m = m - \alpha \cdot \frac{\partial}{\partial m}(MSE)$$

◆ **Intercept update:**

$$b = b - \alpha \cdot \frac{\partial}{\partial b}(MSE)$$

🔗 α (alpha) = **learning rate** (step size)

— Calculate Gradient for b

$$db = -\frac{2}{n} \sum (y - \hat{y})$$

$$\text{sum(error)} = 0.1 + 0.2 - 0.7 + 0.4 = 0$$

🔗 So:

$$db = -\frac{2}{4} \times 0 = 0$$

💧 Final Result

$$dm = 0$$

$$db = 0$$

Update m and b

$$m = m - \alpha \cdot dm$$

$$b = b - \alpha \cdot db$$

🔗 Since gradients are zero:

$$m = 1.9$$

$$b = 0$$